Presentation 2

- Links to presentation(s) and code(s) on GitHub
  - https://app.powerbi.com/links/Yk9C5Wa-lf?ctid=7d76d361-8277-4708-a477-64e8366cd1bc&pbi_source=linkShare (dashboard)
  - https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Code/Dashboard/Queue_Time_Analysis_Oct20_Jake_Miller.ipynb (code)
  - https://docs.google.com/presentation/d/1-LnITFJ8jJQ_EgSvWeso75PoWs314rQuZgU1kCoULfs/edit?slide=id.g33fb61485ab_0_110#slide=id.g33fb61485ab_0_110 (presentation)
- What did you do?
  - Used the CAR dataset to find the duration of time spent in queue across menus
  - Code displays filtering for outliers as well as splitting up queues into high volume and medium volume(only used high volume for this dashboard but should be easily repeatable for medium volume)
- How does it help the project?
  - Goal is to help us understand both the busiest queues and which queues people wait the longest in
  - This can help pinpoint the most important queues so we can assign more or less people to certain queues
- Issues faced (if any)
  - None outside of having too much information to consolidate well into a concise presentation
- Attempts to resolve issues (if any)
  - Push some of what I want to do to next presentation(will talk about in next steps)
- Issues resolved (if any)

- Nothing to be said here

- Next steps
  - Continue this analysis with focus on medium sized queues(code already written for the data prep)
  - Look into callback data
    - Analyze which menus have people more like to pickup and return a callback
    - More answered callbacks means more importance, fewer answered callbacks mean the person does not value being called back and solving their issue

Presentation 1

- Links to presentation(s) and code(s) on GitHub
  - https://app.powerbi.com/links/j1goJgfcFe?ctid=7d76d361-8277-4708-a477-64e8366cd1bc&pbi_source=linkShare
  - https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Code/Dashboard/Jake_Miller_Pres_1_code.ipynb
- What did you do?
  - Used CAR dataset to find the duration of time spend on the ClosedMenu of a caller
  - Found the distribution of Termination Reasons
- How does it help the project?
  - ClosedMenu analysis helps us understand how long people stay on this menu and if they are getting the full information from this menu

    - Termination Reasons will be looked into more to see what parts of calls people terminate calls at and for what reasons
        - E.g. do agents terminate calls more at a certain part of the section vs customers
- New Metric Ideas:
    - I am going to offer a couple new metrics based on what I am exploring(AI for helping with some inspiration to jump off of)
        - Menu End Call Rate(MECR) and/or Menu Leave Rate(MLR)
            - Both these metrics deal with analyzing what a person does in the first N seconds of entering a certain menu
            - MECR will say the percent of people that a call ends within N seconds of answering menu
                - This would mean a full call end
            - MLR will say the percent of people that click a button to leave that menu(but not that whole call) within N seconds
            - Both Metrics can be helpful in analyzing the clarity of each menu part
            - E.g. if one menu sees a high MECR it likely means this is a menu that is commonly revisited and upon a customer hearing the menu again they realize they cant solve their problem and hang up
                - A high MLR would say that customers are likely choosing the first option the most often, in which case further analysis can be done to see if the later options are unnecessary or should be put somewhere else.
            - The N count can be specified based on call menu length, a 3 choice auto response would have a smaller N than an 8 choice response(assuming choice lengths are comparable)
        - This next menu is somewhat a copy of an already used metric but more grouping

        - Call length(CL) and Menu Total(MT) grouped by a specific menu reached in a call
            - If one menu is constantly involved in calls that reach a ton of menus or in calls that are very long, it means that the customers reaching this menu are not getting to the support they need to as efficiently as others
                - In addition, longer call length and lots of menus could point to customer persistence, their want to stay on the line for as long as needed. This could point to a certain menu being involved in calls that customers tend to find more high priority, in which case we can diagnose what this high priority is and make it easier for the customer to reach

- Issues faced (if any)
    - No concrete issues, just a slow grind of understanding the data we are working with
- Attempts to resolve issues (if any)
    - More looking through data and thinking about different ideas
- Issues resolved (if any)
    - Nothing to be said here
- Next steps
    - Look more deeply into termination reasons, I'm mainly interested to see in system disconnects, see where that happens the most or if its random
        - Analyzing this can help us determine what parts of the phone line need to be optimized
            - For example, if system disconnects happen disproportionately when connecting to a certain third party vendor(not sure if we can see each different vendor, in general third party connects works too), that can tell us that there is a problem with the vendor that should be looked into

        - Another example is if customers leave more often on a certain menu item, it likely means when customers get to this line they are not being helped(this can be taken further to if a customer ends a call within 2-3 “transfers” of a menu)

- Further the ClosedMenu Analysis to do ClosedQueueMenu instead
    - On the initial level, it can have the same effect as the ClosedMenu idea where people may not be getting information they need before they hang up
    - Moreso, based on the prior menu selections leading up to the queue, can implement custom messages to point customers in the exact direction they need for information about their problem
        - E.g. customer wants to contact an attorney about divorce, can be pointed to the website that mentions walk in clinics to talk to attorneys, or a customer concerned about eviction can be pointed to the eviction section of the website